Netflix Content Strategy: Data Science Business Insights Report

Auspify Technologies Data Science Internship, Task 6 (Advanced)
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Executive Summary

This report combines exploratory analytics, machine learning and forecasting on a cleaned Netflix titles dataset (8,786 titles) to answer one business question: **how is the catalogue changing, and what should a content-strategy team do about it?**

The five most important findings:

1. **Volume growth has plateaued.** Titles added per year peaked at 2,013 in 2019, fell 6.7% in 2020 and are on pace for roughly 2,000 in 2021.
2. **The catalogue is now majority international.** The non-US share of added titles rose from about 21% in 2014 to between 58% and 68% every year since 2017.
3. **Series are growing but are mostly short-lived.** TV shows rose from 25% of additions (2018) to about 34% (2021), yet 67% of all series have only one season.
4. **The genre mix is shifting.** Between 2017 and 2020, Comedies gained 8.7 percentage points of additions while Documentaries lost 11.1.
5. **Audience group can be predicted from metadata** with about 64% accuracy against a 46% baseline, which is useful for triage and tagging but not accurate enough to run without human review.

Important scope note: the data contains no viewing, cost or revenue figures. Every recommendation therefore concerns *catalogue composition* (what is added, from where, for whom), not viewer demand.

An interactive dashboard accompanies this report, with filters for content type and year range.

1. Objectives and Approach

The project follows an end-to-end data science workflow:

Stage	Work performed
Data preparation (Task 1)	Removed 4 hidden duplicates, converted placeholder text ("Not Given") to Unknown, fixed dates, split duration into minutes and seasons, grouped 14 rating codes into four audience groups
Exploration (Task 2)	Distribution by type, country, genre, rating and time
Recommendation engine (Task 3)	Content-based recommender using TF-IDF and cosine similarity
Trend forecasting (Task 4)	Release-year trends and six forecasting models
Classification (Task 5)	Six classifiers for two problems (Movie vs TV Show, audience group)
Business insights (Task 6)	This report, the analysis notebook and the interactive dashboard

2. Data and Quality

- **Size after cleaning:** 8,786 titles (from 8,790 raw rows).
- **Mix:** 69.7% movies, 30.3% TV shows; 85 producing countries; the United States accounts for 36.9% of titles.
- **Hidden missing values:** 29% of titles have no director and about 3% have no country (stored as placeholder text, not true blanks).
- **Time coverage:** titles were added between 2008 and 25 September 2021. Because 2021 is a partial year, it is flagged wherever it appears.
- **Snapshot bias:** the file shows what was on the platform in 2021, so older titles that have since left are absent.

3. Key Findings

3.1 Catalogue growth has levelled off

Year added	Movies	TV shows	Total	Change
2016	251	175	426	
2017	836	349	1,185	+178%
2018	1,236	411	1,647	+39%
2019	1,422	591	2,013	+22%
2020	1,284	595	1,879	−7%
2021 (to 25 Sep)	993	505	1,498	partial

Growth was explosive until 2019 and then flattened. The 2020 dip coincides with pandemic production disruption, so it should not be over-interpreted as a permanent change.

3.2 The catalogue has become international

The US share of added titles fell from 79% (2014) to 32% (2018) and has since stayed between 39% and 42%. Since 2019 the largest non-US sources are India, the United Kingdom, Pakistan, Japan, Canada, South Korea and France.

Comparing 2019–20 with 2017–18 (when total additions grew 37%): Japan +87%, France +52% and South Korea +46%. **India fell 15%**, even though it is the largest non-US source.

3.3 Series are rising, but most run for one season

TV shows grew from 25% of additions in 2018 to about 34% in 2021. However, 67% of series have a single season, and the figure is still 70% for shows released in 2016 or earlier, so this is not simply because many shows are new.

3.4 Genre and audience mix

- From 2017 to 2020, Comedies gained 8.7 percentage points of additions and Documentaries lost 11.1. (A 2016 starting point gives a different picture because 2016 was a small and unusual year, so this result is indicative.)

- Adult-rated content is steady at roughly 45–47% of additions.
- Kids and Older Kids content is about 23% of the catalogue; its share of additions dipped to roughly 21–22% in 2018–19 and recovered to 23–24% in 2020–21.

3.5 Timing and format

- 28% of additions arrive on a Friday, and October to December each account for about 10% of additions against 7–8% in most other months.
- The median movie runs 98 minutes, and the median gap between release and addition is one year; 55% of titles are added within a year of release.

4. Predictive Models

4.1 Audience classification

A Random Forest predicts the audience group (Kids, Older Kids, Teens, Adults) from genres, country, type, duration and timing features. The rating itself is excluded because it is the answer.

Measure	Result
Test accuracy	63.8%
5-fold cross-validation accuracy	62.4%
Macro F1	0.62
Majority-class baseline	46.0%

The model is strongest for Kids content (F1 about 0.82 in the Task 5 evaluation) and weakest for Teens and Older Kids, which overlap heavily. For the 82 unrated titles it suggests 59 Adults, 15 Teens, 7 Older Kids and 1 Kids; the Adults-heavy output should be reviewed by a person before use.

A companion problem, Movie vs TV Show, reaches about 98.5% accuracy, but this is largely because the catalogue's genre labels and director field differ by type, so it is an easy task rather than a strong claim about content.

4.2 Forecasting titles added per year

Four models were compared with a rolling one-year-ahead test (2015–2020). Damped Holt exponential smoothing was best (RMSE about 287 titles) and beat the naive baseline (about 421).

Year	Forecast	Rough 80% range
2021	about 2,060	1,690 – 2,430
2022	about 2,240	1,720 – 2,760
2023	about 2,420	1,780 – 3,060

The 2021 estimate is consistent with the 1,498 titles added by 25 September (about 73% of the year). The 2022–23 figures carry wide uncertainty, because the model still extrapolates some growth after the 2019 peak; they are best treated as planning scenarios.

5. Business Recommendations

#	Recommendation	Evidence
1	Plan for **flat-to-modest growth of about 1,700–2,400 additions a year** and add quality metrics (completion, retention) beside volume	Plateau after 2019; forecast range
2	**Keep investing in Japan, South Korea and France; review the India pipeline**	Fastest-growing sources vs India −15%
3	Use viewing data to separate planned limited series from cancelled shows, and **prioritise multi-season franchises where retention supports it**	67% of series have one season
4	**Maintain a deliberate genre portfolio**; check whether the comedy-up, documentary-down shift is strategy or licensing availability	+8.7 / −11.1 point shifts

#	Recommendation	Evidence
5	**Keep a family-content floor of about 20–25%** and automate kid-safe tagging	Kids share dip and recovery; best model class
6	Use the audience model as a **triage tool with human review** and improve metadata completeness	64% accuracy; 29% missing directors; 82 unrated titles
7	Align marketing calendars with **Friday and Q4 peaks** and test whether spreading releases reduces competition	28% of additions on Fridays

6. Risks and Limitations

- **No demand data.** Without viewing hours, cost or subscriber retention, the findings cannot say which content performs well.
- **Snapshot and survivorship bias.** The 2021 file omits titles that have left the platform, and recent years may be incomplete.
- **Short time series.** Yearly forecasts rest on roughly a decade of meaningful data; the shaded ranges are rules of thumb, not exact statistical intervals.
- **Partial and unusual years.** 2021 is incomplete and 2020 was affected by the pandemic.
- **Classification labels are constructed.** Audience groups come from a hand-made mapping of rating codes, so the model inherits that choice.

7. Next Steps

1. Obtain viewing, cost and retention data per title to move from catalogue insights to performance insights.
2. Add plot descriptions and cast to improve classification and recommendations, especially for Teens and Older Kids.
3. Refresh the analysis with a more recent snapshot and re-test the forecasts.
4. Pilot the audience model on unrated or newly acquired titles, with human review.

Appendix: Project Deliverables

- `Task1_Data_Cleaning_Preprocessing.ipynb` and `Netflix_Cleaned.csv`

- `Task2_Exploratory_Data_Analysis.ipynb`
- `Task3_Recommendation_System.ipynb`
- `Task4_Trend_Prediction.ipynb`
- `Task5_ML_Classification.ipynb`
- `Task6_Business_Insights_Dashboard.ipynb`
- Interactive dashboard (published page)